Geology Labs

Alveary

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Alt. Science Options

Geology Labs GRADE(S): 11-12

About the Course

Labs/fieldwork are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Geology

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
11-12	Geology Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Geology				
Geology Lessons Nature Walks & Scouting: Grades 9-12	Geology Lessons	Geology Lessons	Geology Lessons Nature Notebook: Grades 9-12	Geology Lessons Geology Labs

Planning & Prep

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LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ View Book List Details

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

(No Subject Supplies Assigned)

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

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How To Approach

Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

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Term 1

WEEK 1	⬚ **60m Geology Labs - Lesson 1** **➔ OBSERVE** Begin observations in the field: use your field notebook, collect specimens when allowed, and take pictures where helpful. Begin building a rock, mineral, and/or fossil collection.	
WEEK 2	⬚ **60m Geology Labs - Lesson 2** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 3	⬚ **60m Geology Labs - Lesson 3** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 4	⬚ **60m Geology Labs - Lesson 4** **➔ OBSERVE** When you are driving around, start noticing where there are exposed rocks in your landscape. Take note of road beds that had to be cut out of the ground. Make observations of what you are seeing in your notebook. Include sketches when you are able. If there are any rivers, creeks, streams, etc. nearby, observe what sort of soil/rock has been exposed in the banks.	
WEEK 5	⬚ **60m Geology Labs - Lesson 5** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 6	⬚ **60m Geology Labs - Lesson 6** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 7	⬚ **60m Geology Labs - Lesson 7** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 8	⬚ **60m Geology Labs - Lesson 8** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 9	⬚ **60m Geology Labs - Lesson 9** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 10	⬚ **60m Geology Labs - Lesson 10** **➔ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 11	⬚ **60m Geology Labs - Lesson 11** **➔ OBSERVE** Continue your local exploration, collection, and documentation.	

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Term 1

WEEK 12	⬚ **60m Geology Labs - Lesson 12** *Exam Week* **→ EXAMS** Answer question(s) related to course.	

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Term 2

WEEK 1	⬚ **60m Geology Labs - Lesson 13** **➔ OBSERVE** Continue to use this time to explore your local geology. Continue sketching, describing, and documenting.	
WEEK 2	⬚ **60m Geology Labs - Lesson 14** **➔ OBSERVE** Continue to use this time to explore your local geology. Continue building a rock, mineral, and/or fossil collection. Continue sketching, describing, and documenting.	
WEEK 3	⬚ **60m Geology Labs - Lesson 15** **➔ OBSERVE** Continue to use this time to explore your local geology. Continue building a rock, mineral, and/or fossil collection. Continue sketching, describing, and documenting.	
WEEK 4	⬚ **60m Geology Labs - Lesson 16** **➔ OBSERVE** Continue to notice where there are exposed rocks in your landscape. Take note of road beds that had to be cut out of the ground. Make observations of what you are seeing in your notebook. Include sketches when you are able. If there are any rivers, creeks, streams, etc. nearby, observe what sort of soil/rock has been exposed in the banks.	
WEEK 5	⬚ **60m Geology Labs - Lesson 17** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 6	⬚ **60m Geology Labs - Lesson 18** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 7	⬚ **60m Geology Labs - Lesson 19** **➔ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 8	⬚ **60m Geology Labs - Lesson 20** **➔ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 9	⬚ **60m Geology Labs - Lesson 21** **➔ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 10	⬚ **60m Geology Labs - Lesson 22** **➔ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 11	⬚ **60m Geology Labs - Lesson 23** **➔ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 12	⬚ **60m Geology Labs - Lesson 24** *Exam Week* **➔ EXAMS** Answer question(s) related to course.	

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Term 3

WEEK 1	⬚ **60m Geology Labs - Lesson 25** **→ OBSERVE** Continue to use this time to explore your local geology.	
WEEK 2	⬚ **60m Geology Labs - Lesson 26** **→ OBSERVE** Continue to use this time to explore your local geology.	
WEEK 3	⬚ **60m Geology Labs - Lesson 27** **→ OBSERVE** Continue to use this time to explore your local geology.	
WEEK 4	⬚ **60m Geology Labs - Lesson 28** **→ OBSERVE** Continue to notice where there are exposed rocks in your landscape. Take note of road beds that had to be cut out of the ground. Make observations of what you are seeing in your notebook. Include sketches when you are able. If there are any rivers, creeks, streams, etc. nearby, observe what sort of soil/rock has been exposed in the banks.	
WEEK 5	⬚ **60m Geology Labs - Lesson 29** **→ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 6	⬚ **60m Geology Labs - Lesson 30** **→ OBSERVE** Continue observations in the field. Sketch, describe, and document your local geology.	
WEEK 7	⬚ **60m Geology Labs - Lesson 31** **→ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 8	⬚ **60m Geology Labs - Lesson 32** **→ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 9	⬚ **60m Geology Labs - Lesson 33** **→ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 10	⬚ **60m Geology Labs - Lesson 34** **→ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 11	⬚ **60m Geology Labs - Lesson 35** **→ OBSERVE** Continue your local exploration, collection, and documentation.	
WEEK 12	⬚ **60m Geology Labs - Lesson 36** *Exam Week* **→ EXAMS** Answer question(s) related to course.	